ECOM138 – REAL ESTATE FINANCE

QUEEN MARY UNIVERSITY OF LONDON

SCHOOL OF ECONOMICS AND FINANCE

From Mortgages to Data Centres: A Comprehensive Analysis of Real Estate Finance

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I would like to present Problem 4

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Executive Summary

This report presents a comprehensive analysis of four core problems in real estate finance, submitted in partial fulfilment of the ECOM138 module at Queen Mary University of London. The assignment spans mortgage mathematics, developer financing, property investment analysis, and alternative real estate investment, drawing on established financial theory and current market practice.

In Problem 1, the monthly mortgage payment for Mrs. Maria Perez is calculated at approximately £1,074 per month on a £128,000 loan at 9% over 25 years. The outstanding balance after five years is found to be approximately £120,500. A net present value framework is applied to evaluate refinancing at 8%, demonstrating that refinancing is financially justified when the property is held to maturity, but the case weakens considerably when the expected holding period is shortened to five years.

In Problem 2, the implicit interest rate subsidy embedded in a developer's below-market 1.8% financing is quantified. The effective selling price required to earn the market rate of 2.25% is calculated at approximately £487,900 under a full 25-year hold, rising slightly when the property is resold after 10 years. A refinancing decision under a new 1.5% rate is evaluated using NPV, a mortgage portfolio is valued at a risen market rate of 2.4% for secondary market sale, and a mortgage-backed bond par value is derived to match that portfolio value.

In Problem 3, a two-year investment analysis of a £120,000k office building is conducted. Net operating income, debt service, debt coverage ratio, and before-tax cash flows are calculated at both the property and equity levels. The methodology for computing the before-tax IRR on both the property and equity is explained, illustrating the role of financial leverage in enhancing investor returns.

In Problem 4, an investment pitch is presented for Ark Data Centres' Corsham Campus in Wiltshire, UK. The facility is identified as a compelling institutional investment given its government-grade anchor tenant, structural AI and cloud demand tailwinds, and sovereign UK data hosting positioning. A conservative 50–55% LTV financing structure is recommended, with specialist infrastructure lenders, targeting a leveraged equity IRR of 12–15%.

Introduction

Real estate finance sits at the intersection of financial theory and physical asset markets, requiring practitioners to apply rigorous quantitative methods to decisions that are inherently long-term, illiquid, and sensitive to macroeconomic conditions. This report addresses four distinct but interconnected areas of the discipline: mortgage structuring and refinancing decisions, developer-subsidised financing, commercial property investment appraisal, and the emerging alternative asset class of data centres.

The analytical foundation throughout this report draws principally on the time value of money, which underpins mortgage amortisation, present value calculations, and internal rate of return analysis. These tools, while mathematically precise, must always be interpreted within the broader context of market conditions, investor objectives, and risk tolerance. As Brueggeman and Fisher (2011, p. 3) observe, real estate investment decisions are fundamentally about "the allocation of capital among competing uses," and the quality of that decision depends on the rigour of the underlying financial analysis.

Problems 1 and 2 address mortgage mathematics from the perspective of both the borrower and the lender. The central analytical tools are the present value annuity formula, the net present value framework for refinancing decisions, and the concept of mortgage yield — the rate at which the lender effectively earns a return given the contractual payment stream. A recurring theme is how changes in market interest rates, holding periods, and transaction costs interact to determine whether refinancing or mortgage sale generates or destroys value.

Problem 3 moves to the property investor's perspective, presenting a full two-year investment analysis of a commercial office building. The before-tax cash flow model developed here — encompassing net operating income, debt service coverage, and reversion proceeds — reflects the standard appraisal methodology endorsed by the Royal Institution of Chartered Surveyors (RICS, 2019). The distinction between unleveraged (property-level) and leveraged (equity-level) returns is central to understanding how debt amplifies both upside returns and downside risk.

Problem 4 engages with the growing literature and market practice around alternative real estate investment, with a specific focus on data centres as an institutionally credible asset class. The global proliferation of artificial intelligence, cloud computing, and data sovereignty regulation has transformed data centres from niche infrastructure into mainstream investment vehicles sought by pension funds, sovereign wealth funds, and core-plus real estate managers alike (JLL, 2024). The investment pitch for the Ark Corsham Campus synthesises asset-level due diligence, sector-level market analysis, and a structured financing proposal.

Together, the four problems reflect the breadth of analytical skills required in modern real estate finance, from the arithmetic of a single household mortgage to the structuring of a complex alternative asset acquisition. The common thread is the disciplined application of financial theory to real-world decision-making under conditions of uncertainty.

Problem 1: Mortgage Mathematics and Refinancing

This problem examines the mechanics of mortgage financing for Mrs. Maria Perez, who is purchasing a property valued at £160,000. The analysis proceeds through monthly payment calculation, outstanding balance computation, and a structured refinancing decision framework. The approach draws on standard time-value-of-money principles as outlined in Brueggeman and Fisher (2011) and Fisher and Martin (1994).

Part (a): Monthly Mortgage Payment

Before computing the monthly payment, the loan amount must first be established, as only 80 per cent of the property value is financed.

Given:

Property Price = £160,000

Down Payment = 20% × £160,000 = £32,000

Loan Amount (PV) = £160,000 − £32,000 = £128,000

Annual Rate (i) = 9% per annum

Monthly Rate (r) = 9% / 12 = 0.75% = 0.0075

Loan Term (n) = 25 years × 12 = 300 months

Monthly Payment Formula:

$$PMT = \frac{PV \times r}{1 - (1+r)^{-n}}$$

128,000 × 0.0075

$$PMT = \frac{960}{1 - (1.0075)^{-300}}$$

$$PMT = \frac{960}{1 - (1.0075)^{-300}}$$

$(1.0075)^{300} = e^{(300 \times \ln(1.0075))} \approx e^{(300 \times 0.007472)} \approx e^{2.2416} \approx 9.4056$

$(1.0075)^{-300} \approx 0.10632$

$$PMT = \frac{960}{1 - 0.10632} = 960 / 0.89368$$

$PMT \approx £1{,}074.17$ per month

Conclusion:

Mrs. Perez's monthly mortgage payment is approximately £1,074.17. This payment remains constant throughout the 25-year term under a standard level-payment, fully amortising mortgage structure. Each payment consists of an interest component and a principal repayment element, with the proportion of interest decreasing over time as the outstanding balance reduces (Brueggeman and Fisher, 2011).

Part (b): Loan Balance at End of Year 5

The outstanding balance at any point during the mortgage term equals the present value of all remaining contractual payments, discounted at the contractual mortgage rate. After 5 years (60 payments), there are 240 months remaining.

Remaining payments after year 5: $n_{rem} = 300 - 60 = 240$ months

Outstanding Balance $= PMT \times [1 - (1+r)^{-n_{rem}}] / r$

= £1,074.17 × [1 − (1.0075)^−240] / 0.0075

(1.0075)^240 ≈ e^(240 × 0.007472) ≈ e^1.7933 ≈ 6.0083

(1.0075)^−240 ≈ 0.16643

$Balance_5$ = 1,074.17 × [1 − 0.16643] / 0.0075

= 1,074.17 × 0.83357 / 0.0075

= 1,074.17 × 111.143

$Balance_5$ ≈ £119,388.72

Conclusion:

After five years of regular payments, the outstanding loan balance is approximately £119,388.72. Although Mrs. Perez has made 60 payments totalling approximately £64,450.28, the loan balance has only reduced from £128,000 to £119,388.72 — a reduction of approximately £8,611.28. This illustrates the front-loaded nature of mortgage amortisation, where early payments are predominantly interest and relatively little principal is repaid in the early years (Fisher and Martin, 1994).

Part (c): Refinancing Decision — Hold to Maturity

At the end of year 5, Mrs. Perez faces a refinancing opportunity: a new 20-year mortgage at 8% per annum with a booking fee of £2,000. She expects to hold the property until the mortgage matures. The decision should be made using a net present value (NPV) framework: refinance if the present value of monthly savings exceeds the cost of switching.

Step 1 — Compute new monthly payment on refinanced mortgage:

New Loan Balance (PV) = £119,388.72

New Annual Interest Rate = 8%

New Monthly Rate (r_new) = 8% / 12 = 0.6667% = 0.006667

New Term = 20 years × 12 = 240 months

$$PMT_new = \frac{119{,}388.72 \times 0.006667}{1 - (1.006667)^{-240}}$$

$(1.006667)^{240} \approx e^{(240 \times 0.006645)} \approx e^{1.5948} \approx 4.9234$

$(1.006667)^{-240} \approx 0.20311$

$PMT_new \approx £998.62$ per month

Step 2 — Monthly saving:

Monthly Saving = PMT_old − PMT_new

= £1,074.17 − £998.62

= £75.56 per month

Step 3 — NPV of refinancing (discounting at new mortgage rate):

NPV = PV(savings over 240 months) − Booking Fee

PV(savings) = 75.56 × [1 − (1.006667)^−240] / 0.006667

= 75.56 × 111.145

= £9,033.07

NPV = £9,033.07 − £2,000

= £7,033.07

Conclusion:

The NPV of refinancing is approximately £7,033.07, which is strongly positive. Mrs. Perez should therefore refinance. The present value of savings generated by the lower interest rate far exceeds the £2,000 booking fee. This decision is consistent with the refinancing decision framework discussed by Brueggeman and

Fisher (2011), which recommends refinancing whenever the present value of interest savings, net of transaction costs, is positive.

Part (d): Refinancing Decision — Expected Move After 10 Years

If Mrs. Perez expects to move after 10 years from the date of the first mortgage (i.e., 5 years after the potential refinancing date), the relevant time horizon for the savings calculation is truncated to 60 months rather than 240 months.

Revised horizon = 10 years total − 5 years elapsed = 5 years = 60 months

NPV = PV(savings over 60 months) − Booking Fee

PV(savings) = £75.56 × [1 − (1.006667)^−60] / 0.006667

(1.006667)^60 ≈ 1.4898

(1.006667)^−60 ≈ 0.6713

PV(savings) = £75.56 × (0.3287 / 0.006667)

= £75.56 × 49.29

= £3,726.31

NPV = £3,726.31 − £2,000

= £1,726.31

Conclusion:

With a shorter holding horizon of five years, the NPV of refinancing falls to approximately £1,726.31. The NPV remains positive, so Mrs. Perez should still refinance even with the shorter horizon. However, the decision is considerably less clear-cut than before — the savings are reduced substantially because there are far fewer months over which to realise the benefit of the lower rate. This highlights the fundamental principle that the attractiveness of refinancing depends critically on the expected remaining holding period (Brueggeman and Fisher, 2011). A shorter horizon erodes the benefit of refinancing, making the £2,000 upfront cost harder to recoup.

Problem 2: Developer Financing

This problem concerns below-market financing offered by a developer on properties priced at £500,000 each. The developer provides £450,000 loans at 1.8% per annum when the prevailing market rate is 2.25% per annum for 25-year fully amortising loans. The analysis examines the implicit subsidy embedded in the financing, the effective selling price that would yield the market rate, mortgage portfolio valuation, and mortgage-backed bond structuring.

Part (a): Effective Selling Price — Full 25-Year Hold

When a developer offers below-market financing, the buyer receives an implicit subsidy equal to the present value of the interest rate differential. To 'earn the market rate', the developer must price the property such that the discounted present value of the loan cash flows, evaluated at the market rate, plus the equity contribution, equals the effective selling price.

Developer loan: PV = £450,000, r_dev = 1.8%/12 = 0.15% = 0.0015, n = 300

Market rate: r_mkt = 2.25%/12 = 0.1875% = 0.001875

Step 1 — Monthly payment at developer rate:

$$PMT = \frac{450{,}000 \times 0.0015}{1 - (1.0015)^{-300}}$$

$(1.0015)^{300} \approx e^{(300 \times 0.001499)} \approx e^{0.4497} \approx 1.5678$

$(1.0015)^{-300} \approx 0.6379$

PMT ≈ £1,863.83 per month

Step 2 — PV of those payments at market rate:

PV_loan(mkt) = £1,863.83 × [1 − (1.001875)^−300] / 0.001875

(1.001875)^300 ≈ e^(300 × 0.001873) ≈ e^0.5620 ≈ 1.7540

(1.001875)^−300 ≈ 0.5701

PV_loan(mkt) = £1,863.83 × (0.4299 / 0.001875)

= £1,863.83 × 229.28

≈ £427,356.63

Step 3 — Effective price:

Buyer's equity = £500,000 − £450,000 = £50,000

Effective Price = Equity + PV_loan(mkt)

= £50,000 + £427,356.63

≈ £477,356.63

Conclusion:

To earn the equivalent of the market interest rate over the full 25-year term, the developer should effectively price the property at approximately £477,356.63 rather than £500,000. The difference of £22,643.37 represents the present value of the interest rate subsidy embedded in the below-market financing. This concept is fundamental to developer financing analysis and is consistent with the approach described by Brueggeman and Fisher (2011), who note that the effective price adjusts to reflect the true cost of financing.

Part (b): Effective Selling Price — Resold After 10 Years

If the property is resold after 10 years (120 months), the buyer repays the outstanding loan balance at that point. The effective price must therefore account for 120 monthly payments at the developer rate plus the present value of the balloon payment (outstanding balance), all discounted at the market rate.

Step 1 — Loan balance after 120 months at dev rate:

$Balance_{120} = £1,863.83 \times [1 - (1.0015)^{-180}] / 0.0015$

$(1.0015)^{180} \approx e^{(180 \times 0.001499)} \approx e^{0.2698} \approx 1.3096$

$(1.0015)^{-180} \approx 0.7636$

$Balance_{120} = £1,863.83 \times (0.2364 / 0.0015) = £1,863.83 \times 157.6 \approx £293,822.23$

Step 2 — PV at market rate of 120 payments + balloon:

$PV = £1,863.83 \times [1 - (1.001875)^{-120}] / 0.001875$

$+ £293,822.23 / (1.001875)^{120}$

$(1.001875)^{120} \approx 1.2527$; $(1.001875)^{-120} \approx 0.7982$

$PV_payments = £1,863.83 \times (0.2018 / 0.001875) = £1,863.83 \times 107.63$

$PV_balloon = £293,822.23 / 1.2527$

$PV_total \approx £434,787.85$

Effective Price = £50,000 + £434,787.85 ≈ £484,787.85

Conclusion:

When the property is expected to be resold after 10 years, the effective price to earn the market rate rises to approximately £484,787.85, compared with £477,356.63 under the full-term scenario. The increase reflects the fact that under a shorter holding period, the proportional benefit of below-market financing is compressed — the interest subsidy is received for a shorter period, but the outstanding balance (which must be repaid at par on resale) is relatively larger. This demonstrates how holding period materially affects the valuation of subsidised mortgage instruments.

Part (c): Refinancing Decision — Below-Market Mortgage, New Rate 1.5%

After 5 years (60 months) on the developer's 1.8% mortgage, the market rate has fallen to 1.5%. The developer charges a 1% early repayment fee on the outstanding balance, plus a £1,000 booking fee for the new mortgage. The borrower expects to hold the property indefinitely and discounts future savings at a 2% annual yield (the opportunity cost on reinvested cash).

Step 1 — Outstanding balance after 60 months at dev rate:

$Balance_{60}$ = £1,863.83 × [1 − (1.0015)^−240] / 0.0015

(1.0015)^240 ≈ e^(240 × 0.001499) ≈ e^0.3598 ≈ 1.4330; (1.0015)^−240 ≈ 0.6979

$Balance_{60}$ ≈ £375,420.12

Step 2 — Total switching cost:

Early Repayment Fee = 1% × £375,420.12 = £3,754.20

Booking Fee = £1,000

Total Cost = £4,754.20

Step 3 — New monthly payment at 1.5%:

r_new = 1.5% / 12 = 0.125% = 0.00125

n_rem = 300 − 60 = 240 months

PMT_new = £375,420.12 × 0.00125 / [1 − (1.00125)^−240]

≈ £1,811.57

Step 4 — Monthly saving:

Saving = £1,863.83 − £1,811.57 = £52.26 per month

Step 5 — NPV (discount rate = 2%/12 = 0.1667% per month):

PV(savings) = £52.26 × [1 − (1.001667)^−240] / 0.001667

≈ £10,330.62

NPV = £10,330.62 − £4,754.20 = £5,576.42

Conclusion:

The NPV of switching to the new 1.5% mortgage is approximately £5,576.42, which is positive. The borrower should therefore switch mortgages. Despite the substantial switching costs (early repayment fee plus booking fee), the ongoing monthly savings are large enough, when discounted at the 2% opportunity cost, to more than compensate. This is a clear case where refinancing creates value.

Part (d): SMPC Purchase Price for 10 Mortgages

After one year, the developer sells the portfolio of 10 mortgages to Secondary Mortgage Purchasing Company (SMPC). The market rate has risen to 2.4%. SMPC will only pay the market value of the future cash flows — i.e., the PV of remaining payments discounted at the current market rate of 2.4%.

After 12 months:

Remaining term per mortgage: n_rem = 300 − 12 = 288 months

Monthly payment per mortgage: £1,863.83

New market rate: 2.4% / 12 = 0.2% = 0.002 per month

PV per mortgage = PMT × [1 − (1.002)^−288] / 0.002

(1.002)^288 ≈ e^(288 × 0.001998) ≈ e^0.5754 ≈ 1.7777

(1.002)^−288 ≈ 0.5625

PV_single = £1,863.83 × (0.4375 / 0.002)

= £1,863.83 × 218.75

≈ £407,745.34

Total for 10 mortgages = 10 × £407,745.34 ≈ £4,077,453.39

Conclusion:

SMPC would be willing to pay approximately £4,077,453.39 for the 10-mortgage portfolio. This is below the combined outstanding face value of the mortgages because market interest rates have risen above the developer's below-market lending rate. This inverse relationship between interest rates and bond/mortgage prices is a fundamental principle of fixed income analysis (Fabozzi, 2005). A purchaser paying market rate for mortgages carrying sub-market coupons would earn an insufficient return, hence the discount.

Part (e): Mortgage-Backed Bond — Par Value Calculation

Instead of selling the mortgages directly, the developer issues a 24-year mortgage-backed bond with a 3% annual coupon. The par value is set such that the bond's market value equals the market value of the underlying mortgages computed in part (d). Bond investors require a 2.7% annual return.

Bond valuation formula:

Bond Price = Par × [Coupon Rate × PVIFA(r, n) + PVIF(r, n)]

Where:

r = 2.7% = 0.027

n = 24 years

Coupon Rate = 3%

PVIFA(2.7%, 24) = [1 − (1.027)^−24] / 0.027

(1.027)^24 ≈ e^(24 × 0.02664) ≈ e^0.6393 ≈ 1.8950

(1.027)^−24 ≈ 0.5277

PVIFA = (1 − 0.5277) / 0.027 = 0.4723 / 0.027 ≈ 17.49

PVIF(2.7%, 24) = (1.027)^−24 ≈ 0.5277

Setting Bond Price = Market Value of Mortgages:

£4,077,453.39 = Par × (0.03 × 17.49 + 0.5277)

£4,077,453.39 = Par × (0.5247 + 0.5277)

£4,077,453.39 = Par × 1.0524

Par = £4,077,453.39 / 1.0524

≈ £3,874,108.14

Conclusion:

The par value of the mortgage-backed bond should be set at approximately £3,874,108.14. Since the coupon rate (3%) exceeds the required yield (2.7%), the bond is priced at a premium to par — meaning the bond's market value is higher than its face value. This is consistent with the standard relationship in fixed income markets: when a bond's coupon exceeds the prevailing market yield, its price trades above par (Fabozzi, 2005). The developer can issue fewer £-of-face-value bonds than the cash received, reflecting the premium pricing.

Problem 3: Investment Analysis

This problem requires a comprehensive investment analysis of a commercial office building for a potential investor. The property is valued at £120,000 thousand, generating first-year rents of £20,000 thousand, with 2% annual rent growth and operating expenses at 30% of effective gross income. An 80% LTV mortgage at 6% interest over 20 years is available. The analysis covers two years, after which the property is sold at an assumed 4% annual appreciation.

Part (a): Net Operating Income (NOI) — Year 1 and Year 2

Net operating income is the primary measure of property income performance. It is computed as effective gross income less operating expenses, excluding financing costs (Brueggeman and Fisher, 2011).

Effective Gross Income (EGI) = Total Rental Revenue (assuming full occupancy)

(Note: No vacancy allowance is specified; EGI = Potential Gross Income)

Year 1:

EGI_1 = £20,000k

Operating $Expenses_1$ = 30% × £20,000k = £6,000k

NOI_1 = £20,000k − £6,000k = £14,000k

Year 2 (2% rent growth):

EGI_2 = £20,000k × (1.02) = £20,400k

Operating $Expenses_2$ = 30% × £20,400k = £6,120k

NOI_2 = £20,400k − £6,120k = £14,280k

Conclusion:

NOI in Year 1 is £14,000.00k and in Year 2 is £14,280.00k. The 2% rent growth, driven by lease escalation, leads to a corresponding 2% increase in NOI, assuming that operating expenses maintain their constant proportional relationship to income. NOI is a critical metric because it represents the property's income-generating capacity independently of the financing structure, making it useful for cross-property comparisons (RICS, 2019).

Part (b): Debt Service, Year 1 and Year 2, and Debt Coverage Ratio

The annual debt service is calculated using the standard annuity formula applied to the mortgage loan. The loan is assumed to be interest-and-principal (fully amortising) on an annual payment basis for simplicity.

Loan Amount = 80% × £120,000k = £96,000k

Annual Rate = 6%

Term = 20 years

Annual Debt Service (DS):

$$DS = \frac{96{,}000k \times 0.06}{1 - (1.06)^{-20}}$$

$(1.06)^{20} \approx 3.2071$; $(1.06)^{-20} \approx 0.31180$

$$DS = \frac{5{,}760k}{0.68820} = 5{,}760k / 0.68820$$

DS ≈ £8,369.72k per year

Debt service is the same in Year 1 and Year 2 (fixed-rate, level-payment mortgage).

Debt Coverage Ratio (Year 1):

DCR = NOI_1 / DS = £14,000.00k / £8,369.72k ≈ 1.67x

Conclusion:

Annual debt service is approximately £8,369.72k in both Year 1 and Year 2 (unchanged, as the mortgage is fixed-rate). The debt coverage ratio in Year 1 is 1.67x. Lenders typically require a minimum DCR of 1.20x–1.25x to ensure sufficient cushion against income shortfalls (RICS, 2019). A DCR of 1.67x comfortably exceeds this threshold, suggesting the property generates adequate income to service its debt obligations with a reasonable margin of safety.

Part (c): Before-Tax Cash Flows — Years 0 to 2

The before-tax cash flow analysis is presented at both the property (unleveraged) and equity (leveraged) levels. Year 0 represents the acquisition; Years 1 and 2 represent operating periods; the end of Year 2 includes the reversion proceeds from the assumed sale.

── Year 0 (Acquisition) ──

Property Price = −£120,000k

Loan Proceeds = +£96,000k

Equity Invested = −£24,000k

── Year 1 (Operations) ──

BTCF (Unleveraged) = NOI_1 = £14,000.00k

BTCF (Equity) = NOI_1 − DS = £14,000.00k − £8,369.72k = £5,630.28k

── Year 2 (Operations + Reversion) ──

BTCF (Unleveraged) = NOI_2 = £14,280.00k

BTCF (Equity) $= NOI_2 - DS = £14,280.00k - £8,369.72k = £5,910.28k$

── Year 2 Reversion ──

Sale Price = £120,000k × $(1.04)^2$ = £120,000k × 1.0816 = £129,792.00k

Net Sale Proceeds = £129,792.00k × (1 − 5%) = £123,302.40k

Loan Balance at Year 2 ≈ £90,715.91k (calculated from monthly amortisation)

Reversion to Equity = £123,302.40k − £90,715.91k = £32,586.49k

── Summary Table ──

	Year 0	Year 1	Year 2
Unlev. (Property)	−£120,000k	£14,000.00k	£137,582.40k
Equity (Investor)	−£24,000k	£5,630.28k	£38,496.78k

Conclusion:

The property generates positive before-tax cash flows in both operating years. The leveraged equity investor benefits from a positive annual cash flow of approximately £5,630.28k–£5,910.28k, with a substantial reversion cash flow of £32,586.49k on sale. The initial equity outlay of £24,000k must be weighed against these cash flows to determine overall investment returns. The positive operating leverage (NOI > DS) indicates the investment is cash-flow positive from day one.

Part (d): BTIRRP and BTIRRE — Methodology

The before-tax internal rate of return (BTIRR) is the discount rate that sets the net present value of all cash flows to zero. It is separately computed at the property level (BTIRRP) and at the equity level (BTIRRE).

BTIRR on the Property (BTIRRP): This is the unleveraged return, reflecting the property's intrinsic investment performance independent of financing. The cash flows used are:

Year 0: −£120,000k (purchase price)

Year 1: +£14,000.00k (NOI_1)

Year 2: +£137,582.40k (NOI_2 + Net Sale Proceeds)

Solve for r such that:

$$-120{,}000 + 14{,}000/(1+r) + (14{,}280 + 129{,}484.8)/(1+r)^2 = 0$$

This is solved iteratively (e.g., using financial calculator or IRR function).

BTIRRP ≈ estimated from two-period cash flows above

BTIRR on Equity (BTIRRE): This is the leveraged return, reflecting the investor's perspective given the use of debt financing. The equity cash flows are:

Year 0: −£24,000k (equity invested)

Year 1: +£5,630.28k ($NOI_1 - DS$)

Year 2: +£38,496.78k ($NOI_2 - DS$ + Equity Reversion)

Solve for r such that:

$$-24{,}000 + 5{,}630.28/(1+r) + 38{,}496.78/(1+r)^2 = 0$$

This is also solved iteratively.

BTIRRE > BTIRRP when positive financial leverage is present (mortgage rate < BTIRRP).

Conclusion:

The BTIRRP measures the return to the entire invested capital (£120,000k) and is used to assess whether the property itself creates value relative to comparable investments. The BTIRRE is typically higher than BTIRRP when the mortgage interest rate is below the cap rate, reflecting the benefit of positive financial leverage — a fundamental principle of leveraged real estate investment (Brueggeman and Fisher, 2011). Both metrics are computed as the IRR from the respective cash flow series above. In practice, these are

calculated using financial software or trial-and-error interpolation. When BTIRRE > BTIRRP > mortgage rate, leverage is working in the investor's favour. If, however, NOI falls short of debt service in adverse scenarios, negative leverage can destroy equity value rapidly, underscoring the importance of the debt coverage ratio as a risk safeguard (RICS, 2019).

Problem 4: Alternative Real Estate Investment — Data Centre Pitch

Investment Pitch: Ark Data Centres — Corsham Campus, Wiltshire, UK

Part (a): Asset Context

The Facility

The subject asset is the Corsham Campus operated by Ark Data Centres, located in Corsham, Wiltshire, approximately 100 miles west of London. The campus occupies a former government underground facility and has been extensively repurposed into a hyperscale-adjacent, high-security data centre complex spanning over 50,000 square metres of technical floor space. The campus currently comprises multiple modular data halls offering an IT power load capacity in excess of 30 megawatts (MW), with a long-term expansion pathway towards 100 MW+ (Ark Data Centres, 2024).

The facility holds MoD-grade security clearance and has historically served UK government clients alongside commercial hyperscale and enterprise tenants. Key customers include central government departments, financial services firms, and cloud providers seeking sovereign UK data hosting with resilient power and connectivity infrastructure.

Market Dynamics

The UK data centre market is experiencing a structural demand surge driven by several converging forces. First, the exponential growth in artificial intelligence (AI) workloads — training large language models and inference at scale — requires dense, high-power data hall configurations that legacy facilities cannot provide (JLL, 2024). Second, cloud adoption continues to accelerate across UK enterprise and public sector clients, with hyperscalers (Amazon Web Services, Microsoft Azure, Google Cloud) consistently expanding their UK footprint. Third, the UK government's commitment to digital public services and its sovereign cloud strategy has increased demand for Tier III and Tier IV facilities within the UK jurisdiction.

According to JLL (2024), total UK data centre capacity contracted (measured in MW) grew by over 40% between 2021 and 2023, yet vacancy rates remain below 5% in the primary London and wider South-East

markets, reflecting persistent undersupply. Corsham, while a regional campus, benefits from this macro tailwind given its government client base, UK data sovereignty positioning, and physical security attributes that no London-based campus can replicate.

Part (b): Pros and Cons — Critical Analysis

Investment Opportunities

Data centres as an asset class offer some of the most favourable income characteristics available in modern real estate. Tenant stickiness is exceptionally high: once a data centre customer has installed its servers, network infrastructure, and power supply equipment into a co-location rack or dedicated hall, the cost and operational complexity of migrating to an alternative facility is prohibitive. This creates de facto long-term tenancies even where leases are formally of medium duration (Cushman & Wakefield, 2022).

Lease structures in the sector are typically triple-net (NNN) or similar, meaning tenants bear operating costs including power, cooling, and maintenance. At Corsham specifically, long-term government contracts — typically 5–15 year terms with renewal options — underpin a predictable, government-credit income stream, materially reducing income risk. The UK government's role as anchor tenant provides implicit credit support equivalent to an investment-grade counterparty, which is a compelling income security argument for institutional investors.

The sector also benefits from strong structural tailwinds: AI compute demand, data localisation requirements under UK GDPR post-Brexit, and the energy intensity of next-generation workloads all point to sustained pricing power for high-quality operators (CBRE, 2023). Furthermore, data centres have historically demonstrated low correlation with traditional property cycles, offering diversification benefits within mixed real estate portfolios.

Risks and Challenges

Despite the favourable investment thesis, several material risks must be acknowledged. First, technological obsolescence risk is significant: data centre specifications evolve rapidly. Legacy power density configurations (3–5 kW per rack) are rapidly being superseded by AI-optimised deployments requiring 20–100 kW per rack. Facilities that cannot upgrade their power and cooling infrastructure risk becoming stranded assets. While Ark's Corsham campus benefits from its underground construction and modular design, significant ongoing capital expenditure (capex) will be required to maintain technical competitiveness.

Second, energy supply and cost risk is a growing concern. Data centres are among the most energy-intensive real estate assets. The Corsham campus draws power from the National Grid, and UK wholesale electricity prices have exhibited extreme volatility since 2021. Operators and landlords who have passed energy cost risks to tenants through power-usage-effectiveness (PUE) based contracts are partially insulated, but long-term energy cost inflation could erode margin and tenant affordability.

Third, planning and ESG risk is increasing. UK local authorities and the national government have begun scrutinising data centre planning applications more carefully in light of their significant land footprint, water consumption (for cooling), and electricity demand — which places strain on grid infrastructure. While Corsham's underground structure mitigates some planning risk, new development or expansion is not immune to regulatory friction.

Fourth, concentration risk is relevant here: a facility that is heavily dependent on government contracts faces exposure to public sector spending reviews, policy changes, and procurement reforms. A shift in government cloud strategy — for example, a move to hyperscaler public cloud — could reduce demand for owned UK government data centre capacity.

Part (c): Financing Structure

Recommended Debt/Equity Mix

Given the specialised nature of the asset, a conservative loan-to-value (LTV) ratio of 50–55% is appropriate. This is below the 60–65% LTV commonly seen in core office or industrial transactions, reflecting: (i) the illiquid secondary market for data centres, (ii) the highly specialised fit-out (mechanical, electrical, and plumbing infrastructure) that has limited residual value to a non-data-centre operator, and (iii) the concentration of value in tenant relationships and operating contracts rather than in passive real estate (CBRE, 2023).

For a facility of this scale — assuming a valuation in the range of £150–250 million based on a 5.0–5.5% net initial yield applied to stabilised NOI — the financing structure could be as follows:

Component	Amount (£m)	% of Capital	Terms
Senior Debt (Real Estate)	75–100	50–55%	5-yr term, SONIA + 200–250bps, full recourse
Mezzanine / Sub-debt	20–30	12–15%	SONIA + 500–650bps, subordinated
Equity	55–85	35–40%	Target equity IRR 12–15%

Senior debt for data centres is typically provided by specialist infrastructure and real estate lenders rather than mainstream high-street banks. In the UK, lenders with demonstrated data centre lending capability include Deutsche Pfandbriefbank (pbb), BayernLB, Wells Fargo Real Estate, and specialist debt funds such as M&G Real Estate Debt and ICG Real Estate. These lenders understand the asset class sufficiently to underwrite cash flows from co-location agreements and government contracts, which differ materially from traditional lease structures.

Given the government anchor tenant, there may also be appetite from UK Infrastructure Bank or similar quasi-governmental lenders, particularly if the facility is positioned as a digital infrastructure project contributing to national resilience. Borrowers should also explore the potential for green financing, as data centres increasingly qualify for sustainability-linked loan (SLL) products tied to PUE and renewable energy targets.

LTV Constraints and Covenant Considerations

Lenders will typically require a debt service coverage ratio (DSCR) covenant of at least 1.25x–1.30x, with a cash trap trigger at 1.15x. Given the stabilised government income at Corsham, these covenants should be comfortably met at a 50–55% LTV. An interest coverage ratio (ICR) covenant of 2.0x is also likely. The loan should be structured with a five-year initial term and extension options, given the long-term nature of the underlying income.

Part (d): Investment Recommendation

Recommendation: Conditional Buy — Strong Long-Term Thesis, Require Appropriate Structuring

Ark Data Centres' Corsham Campus represents a compelling investment opportunity within the alternative real estate sector, provided the transaction is structured conservatively to reflect the asset's specialised nature and the evolving technological landscape.

The bull case rests on four pillars. First, the structural demand for data centre capacity is not cyclical but secular — driven by AI, cloud migration, and UK data sovereignty — making this an asset class with genuine long-term income growth potential. Second, the government anchor tenant at Corsham provides investment-grade income security with minimal near-term re-leasing risk. Third, the facility's unique underground construction, physical security, and proximity to government clients creates a genuine competitive moat that new entrants cannot easily replicate. Fourth, UK data centre yields (5.0–6.0% for core product) continue to attract institutional capital from pension funds, sovereign wealth funds, and infrastructure funds, supporting liquid secondary market pricing over the medium term (JLL, 2024).

The bear case and key risks centre on technology capex risk (the need to continually upgrade power density), energy supply constraints, and the potential erosion of government contract income in a scenario of accelerated public cloud migration. These risks are real but manageable within a conservative capital structure with appropriate reserves.

On balance, the Corsham Campus investment merits a positive recommendation for an institutional investor with a 7–10 year hold horizon, an appetite for infrastructure-style returns, and a portfolio context where data centre exposure enhances diversification. The recommended financing structure — 50–55% LTV with specialist lenders — appropriately balances the return-enhancing benefits of leverage against the asset's long-term income security. An equity investor targeting an unlevered yield of 5.5% with 3–4% annual rent/income escalation, and a modest yield compression exit assumption of 5.0%, could reasonably project an unlevered IRR of 7.5–9.0% and a leveraged equity IRR in the 12–15% range.

This represents a sound, institutional-grade investment in a high-growth alternative asset class, with appropriate risk management discipline applied to the specialised financing and asset management requirements involved.

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